

Submission 1

Integrated Navigation for Aircraft: Kalman Filter Design, Bias Estimation, and Fault Tracking

0 Introduction

Multisensor integration is critical for accurate aircraft navigation, combining GPS, IMU, and air-data sensors to estimate the full state vector. This assignment focuses on designing an Extended Kalman Filter (EKF) for state estimation, addressing challenges such as sensor biases and additive faults. Tasks 1–3 cover EKF implementation, bias estimation using an augmented state model, and fault tracking via a random walk model.

1 Kalman Filter Design

1.1 Jacobian Derivation

The Extended Kalman Filter (EKF) requires Jacobian matrices to linearise the nonlinear aircraft dynamics. Three key matrices are derived: $F = \partial f / \partial x$ captures how state errors propagate, $G = \partial f / \partial \omega$ describes process noise coupling, and $H = \partial h / \partial x$ relates state estimates to measurements. These were computed symbolically using MATLAB's Symbolic Math Toolbox. Table 1 presents the six required matrix elements, showing sensitivities such as yaw rate coupling into lateral velocity dynamics and angle-of-attack dependence on body-frame velocities.

Table 1: Required Jacobian matrix elements

Element	Meaning	Expression
$F_{5,4}$	$\dot{V}_v$ to V_u	$-r$
$F_{6,7}$	$\dot{V}_w$ to ϕ	$-g \cos \theta \sin \phi$
$G_{4,6}$	Noise in $\dot{V}_v$	$-V_v$
$G_{9,5}$	Noise in $\dot{\psi}$	$-\frac{\sin \phi}{\cos \theta}$
$H_{7,7}$	GPS roll	1
$H_{11,4}$	AoA to V_u	$\frac{-V_w}{V_u^2 + V_w^2}$

1.2 Kalman Filter

The nonlinear kinematics $\dot{x} = f(x, u) + g(x, u, \omega)$ with measurements $y = h(x) + \nu$ are handled by a five-step DT EKF. At each sample the filter: (1) *predicts* the state forward using `ode45` on the full nonlinear model f ; (2) propagates the error covariance through the linearised, discretised system matrices Φ_{k-1}, Γ_{k-1} obtained via `c2d`; (3) computes the *Kalman gain* K_k from the linearised observation matrix H_k and the innovation covariance $V_e = H_k P_{k|k-1} H_k^\top + R$; (4) *corrects* the state using the nonlinear predicted measurement $h(\hat{x}_{k|k-1})$ and the actual measurement y_k ; and (5) updates the covariance via the Joseph form $P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^\top + K_k R K_k^\top$ to guarantee positive semi-definiteness.

1.3 EKF Implementation and Results

The initial state $\hat{x}_0$ is pulled from the first measurements and wind components are zero. The process noise Q uses IMU accelerometer standard deviations of 0.01 m/s^2 and gyroscope standard deviations of 0.01 deg/s . The measurement noise with GPS position has uncertainties of 5 m (horizontal) / 10 m (vertical), GPS velocity 0.1 m/s , GPS attitude 0.1 deg , TAS 0.5 m/s , and air-data angles 0.5 deg . The initial error covariance P_0 is set to cover known measurement uncertainty in each state.

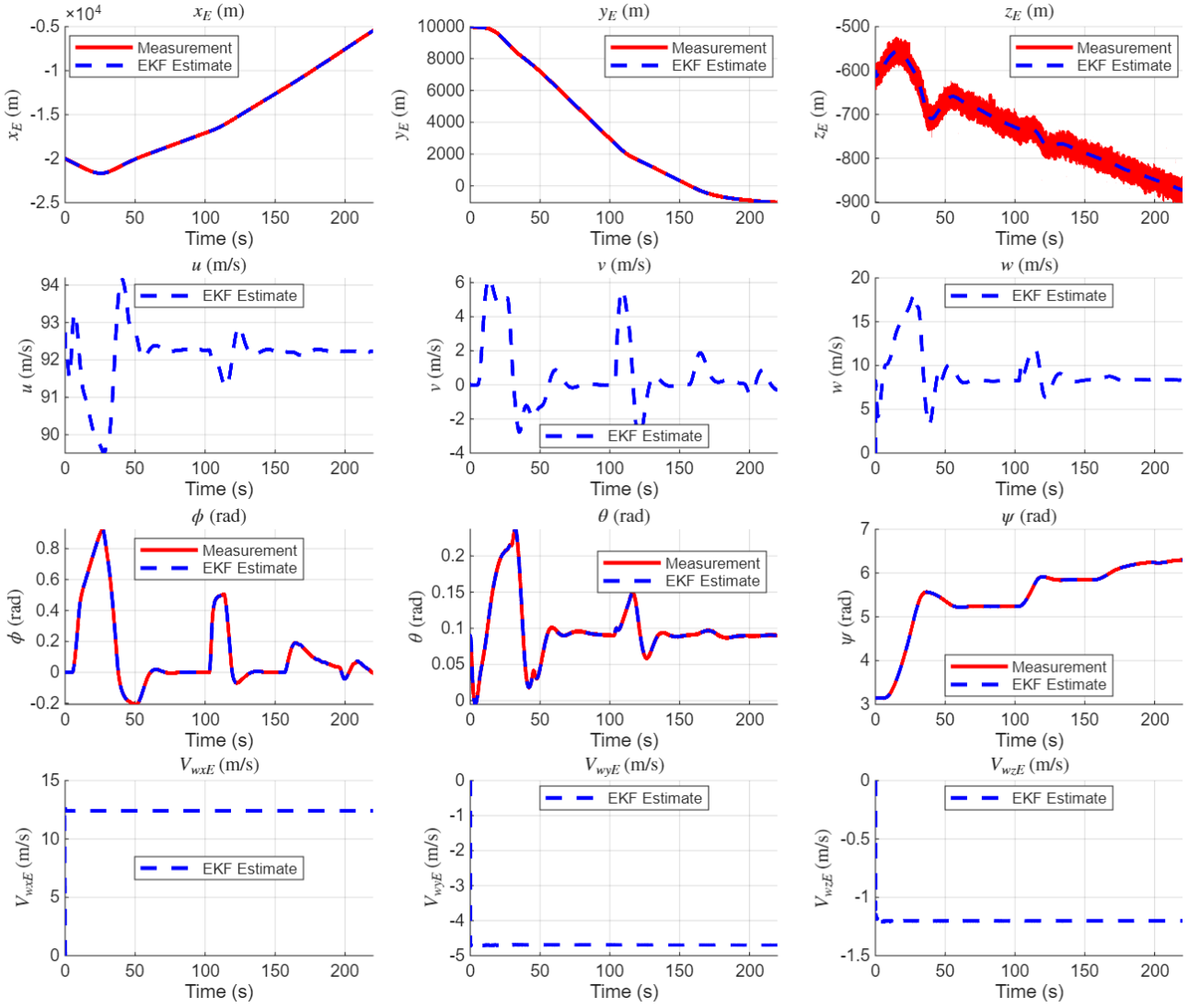


Figure 1: All 12 EKF estimated state trajectories from `dataTask1`. Position (x_E , y_E , z_E) and attitude (ϕ , θ , ψ) are overlaid with GPS measurements (red solid); body velocity (u , v , w) and wind (V_{wxE} , V_{wyE} , V_{wzE}) have no direct measurement.

Figure 1 presents the estimated trajectories for all 12 states. The position states (x_E , y_E , z_E) track the GPS measurements closely; the altitude z_E exhibits noticeably more measurement scatter, consistent with $\sigma_{z_E} = 10$ m versus 5 m for the horizontal channels, while the EKF smooths through this noise effectively. The attitude states (ϕ , θ , ψ) converge within the first few seconds and follow dynamic roll manoeuvres around $t = 20$ – 50 s without visible lag, confirming adequate filter bandwidth. Body-velocity states (u , v , w) have no direct measurement but are constrained indirectly through the nonlinear observation model linking them to GPS earth-frame velocity, TAS, angle of attack α , and sideslip β ; all three settle to physically plausible trajectories with u dominating as expected for forward flight. The wind states (V_{wxE} , V_{wyE} , V_{wzE}) are entirely unobserved and converge the slowest, as the filter must infer them from the residual between predicted and measured earth-frame velocities. They stabilise to approximately constant values, which is consistent with the zero-derivative wind model assumed in the process dynamics.

2 Estimating and Removing Sensor Biases

IMUs are subject to systematic errors that persist beyond noise, the most common being sensor bias. For rate gyroscopes, bias (b_p, b_q, b_r) adds directly to the measured angular rates $p_m = p + b_p + \omega_p$, etc., so every attitude prediction drifts away from the true attitude. GPS attitude measurements supply periodic corrections, but when the filter has no bias state to absorb the offset, it reaches a biased steady-state compromise between the corrupted gyro prediction and the GPS correction signal.

2.1 Effect of Biases on 12-State Filter

The same EKF designed in Task 1 was re-run on `dataTask2.mat`, which contains the same flight trajectory with constant IMU gyroscope biases added to p_m, q_m , and r_m . The filter has no knowledge of these biases, so it integrates biased rates directly into the attitude predictions at every time step.

Figure 2 shows the comparison for the three attitude states (ϕ, θ, ψ) which are the states with the most pronounced degradation. As expected, the biased data estimates diverge from the clean data estimates, with visible steady-state offsets.

The filter can't fully correct for the bias because the model does not include bias states. This poses a significant issue, resulting in a systematic offset primarily in roll and yaw, which propagates into the body-velocity estimates through the rotation matrix coupling in the kinematic equations.

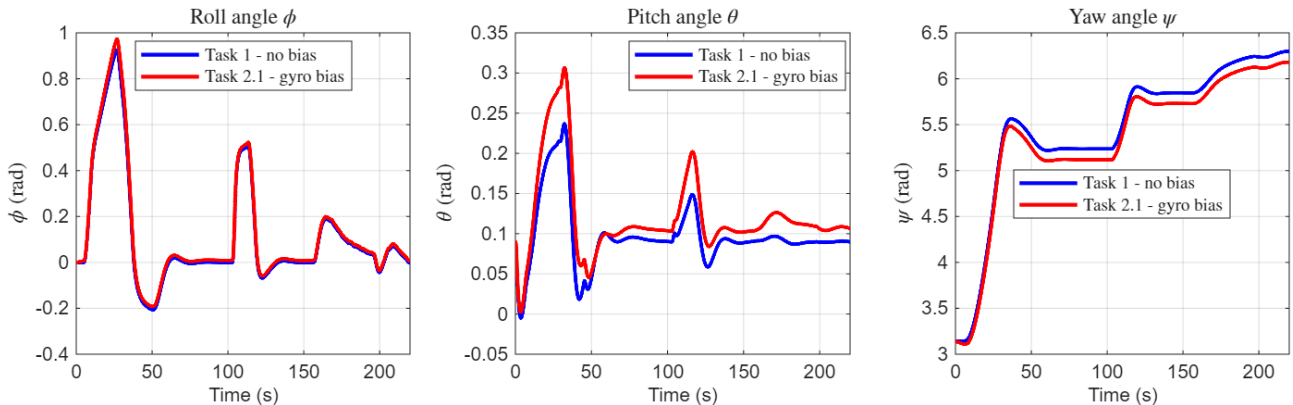


Figure 2: Comparison of EKF attitude state estimates from clean data (blue) to estimates on the biased data (red).

2.2 18-State Bias Estimation

To compensate for constant IMU biases the 12-state model is augmented with six additional states representing the accelerometer biases ($b_{A_x}, b_{A_y}, b_{A_z}$) and the gyroscope biases (b_p, b_q, b_r), giving an 18-state vector $\hat{x} = [x_E, \dots, \psi, b_{A_x}, \dots, b_r, V_{wxE}, \dots, V_{wzE}]^T$. Each bias state is modelled with zero-derivative dynamics $\dot{b} = 0$, so the filter treats them as slowly varying constants to be estimated from the data. The bias-corrected inputs $(A_x - b_{A_x}), (p - b_p)$, etc. replace the raw IMU readings throughout the dynamics f and noise coupling g , while the observation model h remains unchanged.

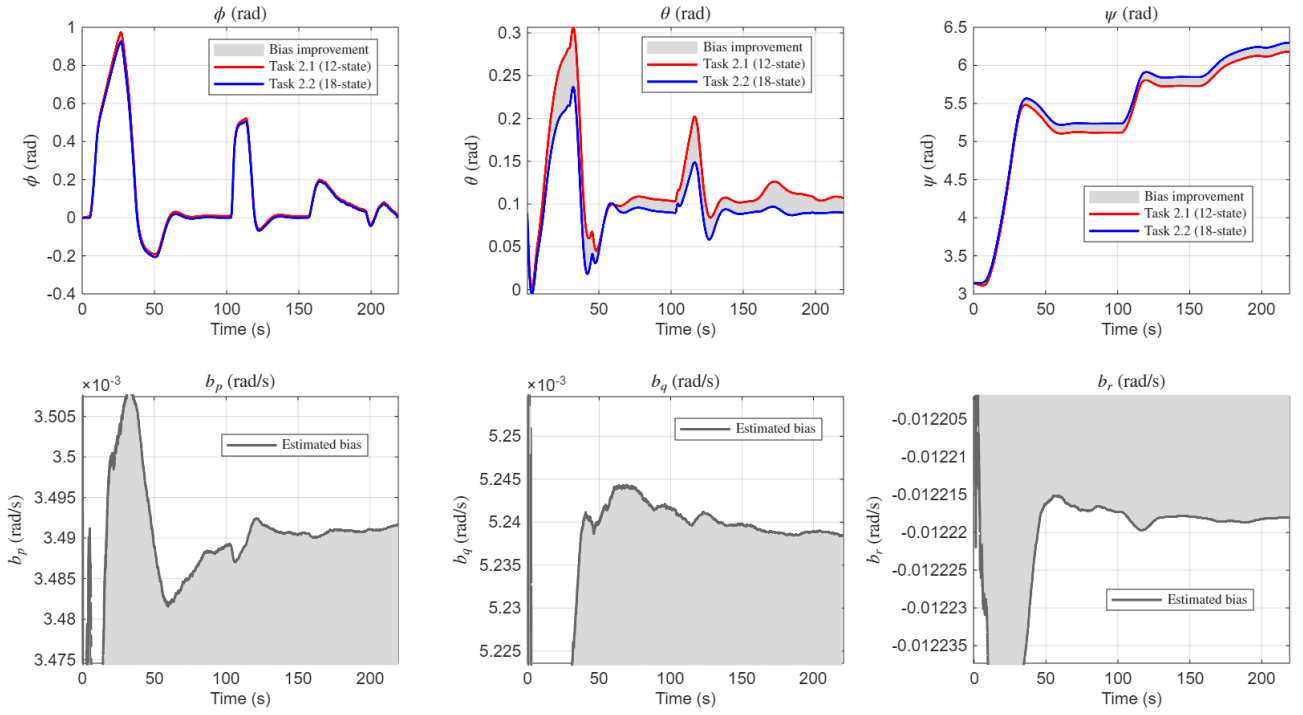


Figure 3: 18-state bias compensation results on `dataTask2`. Top row: attitude estimates from the 12-state filter (red) and 18-state filter (blue), with the grey shaded region highlighting the correction achieved by bias augmentation. Bottom row: estimated gyroscope bias terms (b_p , b_q , b_r) with axes scaled to the converged region; the shaded area shows the deviation from zero.

Figure 3 confirms that the augmented filter effectively fixes the bias-induced attitude errors that were seen in Task 2.1. The top row shows that the 18 state attitude estimates (blue) accurately recover the trajectory that the 12-state filter produced on clean data, with the grey shaded region quantifying the improvement over the uncompensated 12-state estimates (red) on the biased dataset. The bottom three plots present the behaviour of the gyroscope bias estimates, each settling to a constant over time, which is consistent with the biases injected into `dataTask2`. The convergence is enabled by the GPS angle measurements, which provide a solid reference signal that the filter can use to isolate the gyroscope offsets, and compensate for in the bias terms.

3 Estimating Additive Faults

Sensors also tend to experience faults that are not constant but vary over time, like GPS that drifts or steps due to interference. The constant-bias model from Task 2 can't track this, so this section explores how the innovation can be used to detect faults and then extends the EKF with a random walk model to estimate and compensate for them.

3.1 Fault Observation via Innovation

The 18-state EKF from Task 2 (constant-bias model, $\dot{b} = 0$) was re-run on `dataTask3.mat`, which contains sensor faults that vary over time. Because the filter assumes constant biases, it can't track time-varying faults, so their influence appears directly in the innovation.

The innovation $\nu_k = y_k - h(\hat{x}_{k|k-1})$ is the difference between the actual measurement and the filter's prediction. Normally, the innovation is just white noise with covariance $V_e = H_k P_{k|k-1} H_k^T + R$. When a fault corrupts an input, the filter's prediction diverges from the

measurements, causing a systematic shift or spike in the innovation. This makes the innovation a good fault indicator as any deviation from its nominal statistics signals the presence of un-modelled dynamics like a sensor fault.

To locate the first fault instance programmatically, each innovation channel was normalised by its pre-fault standard deviation (computed over the first 5% of the data, after initial convergence). The first time any channel exceeded a 5σ threshold it was flagged as a fault. The GPS velocity channels (ν_{u_E} , ν_{v_E} , ν_{w_E}) had the first detected fault as seen in Figure 4.

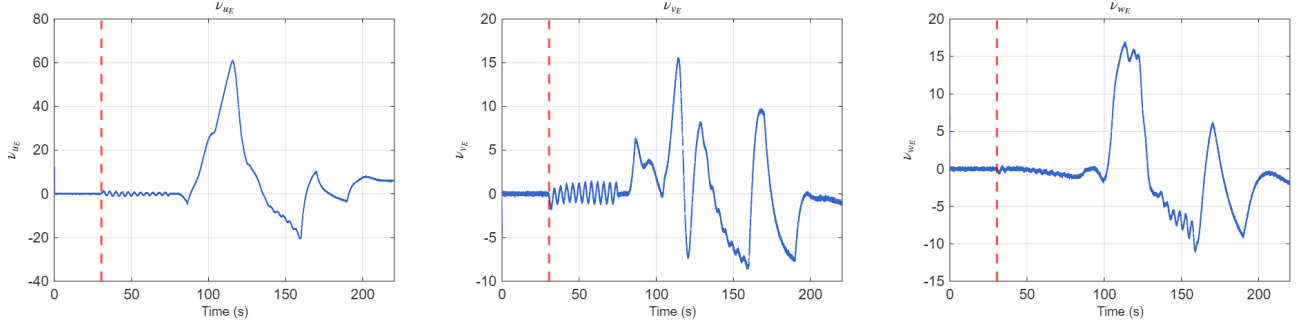


Figure 4: Innovation history of the GPS velocity channels (ground speed components u_E , v_E , w_E) from the 18-state EKF on `dataTask3`. The red dashed line marks the first detected fault instance.

Figure 4 shows the innovation history for the GPS velocity measurements. Before the red dashed line, the innovation fluctuates around zero as expected. After the fault, all three velocity channels show clear deviations away from the initial behaviour. This confirms that the constant-bias model ($\dot{b} = 0$) is insufficient to absorb the error in these faulty signals.

3.2 Random Walk Fault Model

In Task 2, the bias states were modelled with zero-derivative dynamics $\dot{b} = 0$, treating them as unknown constants. For time-varying faults this is inadequate, so the bias dynamics are replaced with a random walk:

$$b_i(t_{k+1}) = b_i(t_k) + \omega_{b_i}(t_k), \quad i \in \{A_x, A_y, A_z, p, q, r\}. \quad (1)$$

This discrete-time relationship is converted to continuous-time for use in the existing linearisation framework as $\dot{b}_i = \frac{1}{T_s} \omega_{b_i}$, where T_s is the sampling period. The process noise vector is therefore extended from 6 to 12 elements: $\omega = [\omega_{A_x}, \dots, \omega_r, \omega_{b_{A_x}}, \dots, \omega_{b_r}]^\top$, and the noise coupling matrix G gains six additional columns corresponding to $\frac{1}{T_s}$ entries on the bias rows. Q is extended to 12×12 accordingly.

The standard deviation σ_b of the fault noise terms controls how quickly the filter can track changes in the fault signals, and introduces the key trade-off:

- **Too large** σ_b : the filter trusts the fault state too little, allowing it to change rapidly at every step. The estimated fault becomes noisy and tracks measurement noise rather than the true fault, making the navigation state estimates worse.
- **Too small** σ_b : the filter treats the fault as nearly constant (approaching the $\dot{b} = 0$ model from Task 2), so it behaves basically the same as it did before, and doesn't mitigate the faults at all.

Through a few iterations, the final values were set to $\sigma_{b_{A_x}} = \sigma_{b_{A_y}} = \sigma_{b_{A_z}} = 0.05 \text{ m/s}^2$ for the accelerometer channels and $\sigma_{b_p} = \sigma_{b_q} = \sigma_{b_r} = 0.5 \text{ deg/s}$ for the gyroscope channels. These values allow the filter to track the fault features present in `dataTask3` whilst keeping the innovation close to white noise.

3.3 Fault Estimation Results

Figure 5 presents the six estimated fault signals (b_{A_x} , b_{A_y} , b_{A_z} , b_p , b_q , b_r) from the random walk EKF applied to `dataTask3`. Axes are scaled to the post-convergence region so that the fault features are clearly visible. Several channels show step-like or ramp-like fault profiles that appear at different times during the flight, consistent with multiple sensor faults being injected into the dataset.

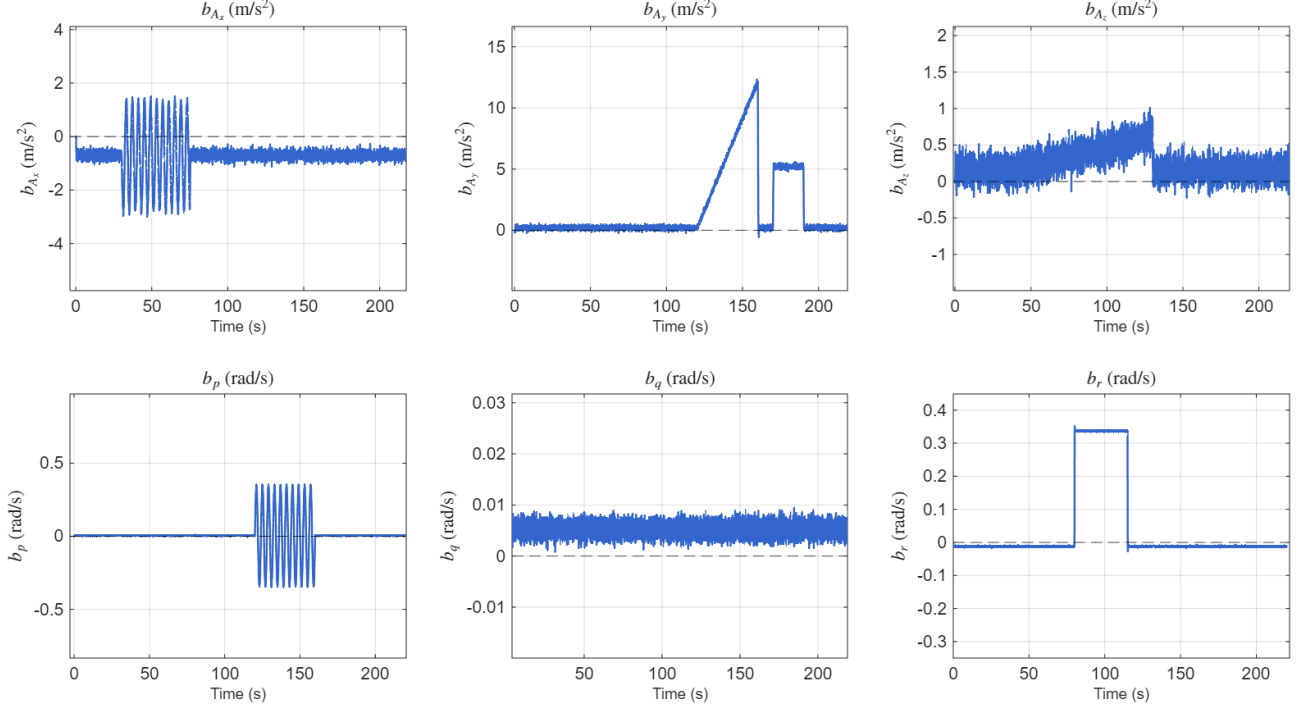


Figure 5: Estimated fault signals from the 18-state random walk EKF on `dataTask3`. Axes are scaled to the converged region to show fault features rather than initial transients.

Figure 6 compares the innovation of the GPS velocity channels before and after fault compensation. The top row reproduces the constant-bias innovation from Task 3.1, where large systematic deviations are visible after the fault onset. The bottom row shows the same channels from the random walk filter. The innovation has returned to approximately zero-mean white noise, confirming that the random walk model successfully absorbs the time-varying fault signals. This shows that the fault estimation framework restores nominal filter performance despite the presence of multiple sensor faults.

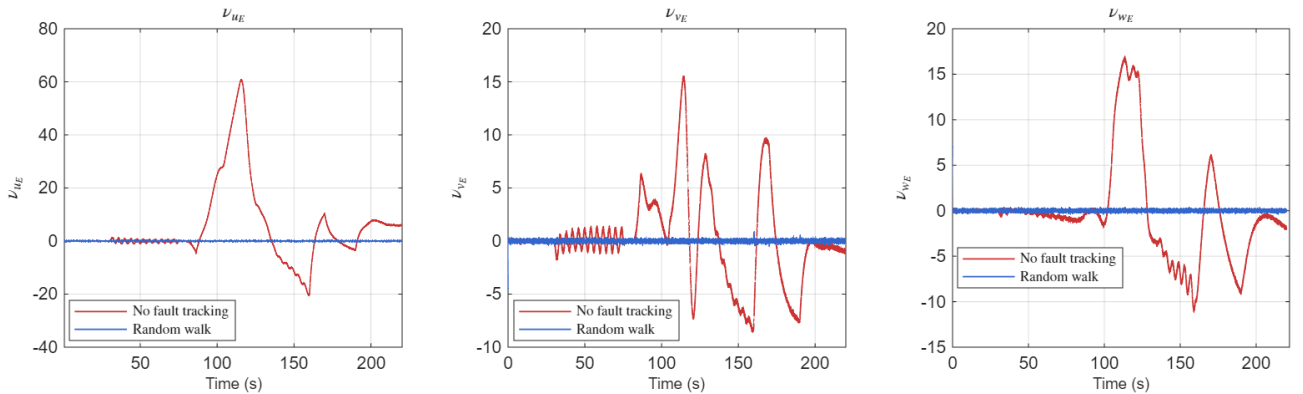


Figure 6: Innovation comparison for GPS velocity channels. The top row shows the constant-bias model from Task 3.1, while the bottom row shows the random walk fault tracking model.